MASTER OF BUSINESS ADMINISTRATION

SEMESTER 3

DADS304

VISUALIZATION

Unit 13

Merging and Appending Data Sources

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1. INTRODUCTION

In Unit 12 we have discussed about connecting to various data sources. Now let's understand the concept of merging of data from different data sources into a single data source in order to visualize the result.

Consider the two sheets as patient master data and patient experience data connected from the file data and xml data, to merge these we need to do right click on the data model and select the merge option, and then we need to choose the sheet or the data source which is required to be merged, in this case we need to merge is patient experience data, for this we need to follow the steps as shown below.

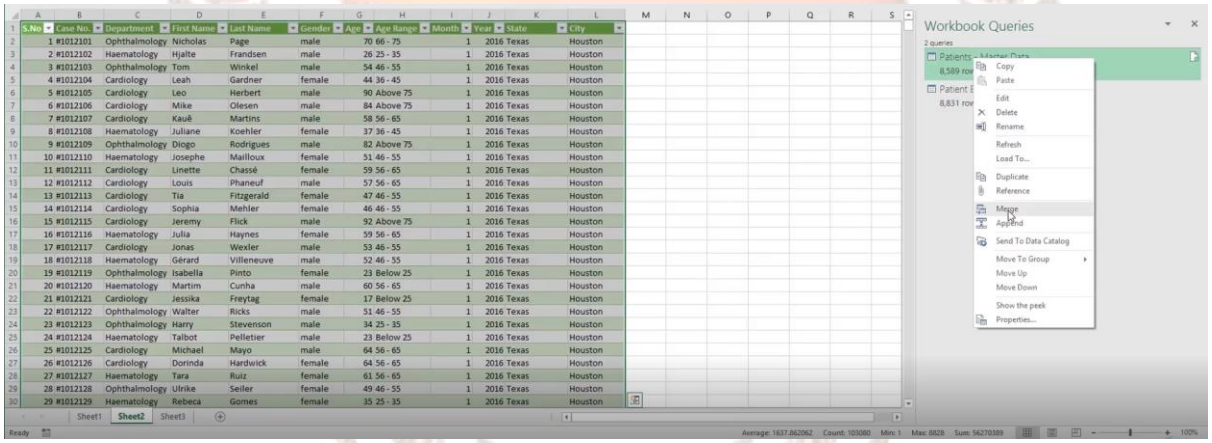


Fig 1: Step 1 Merge Data

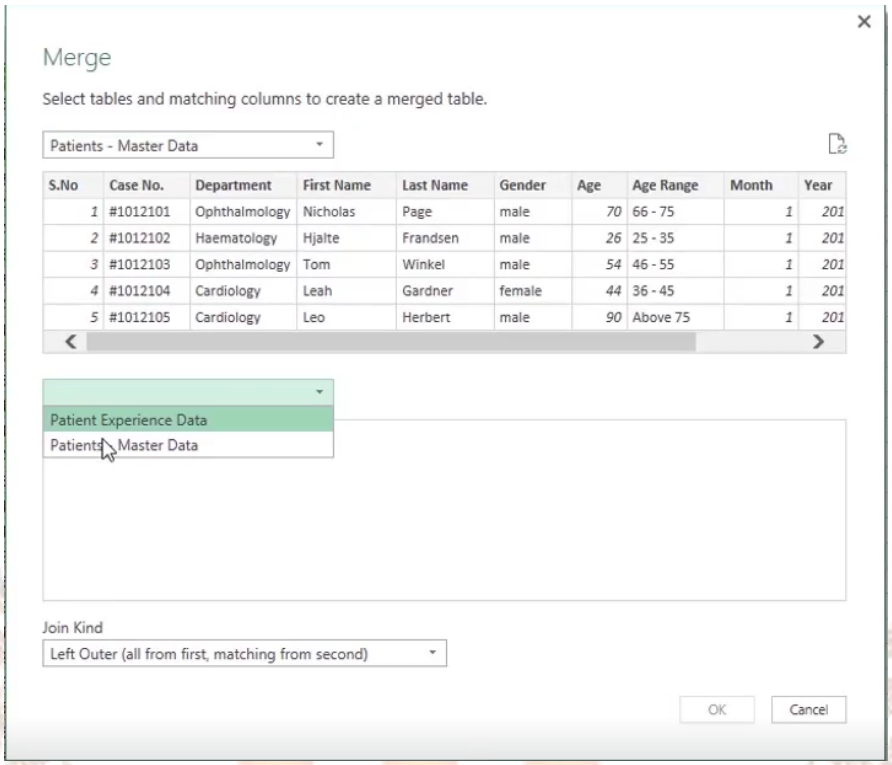
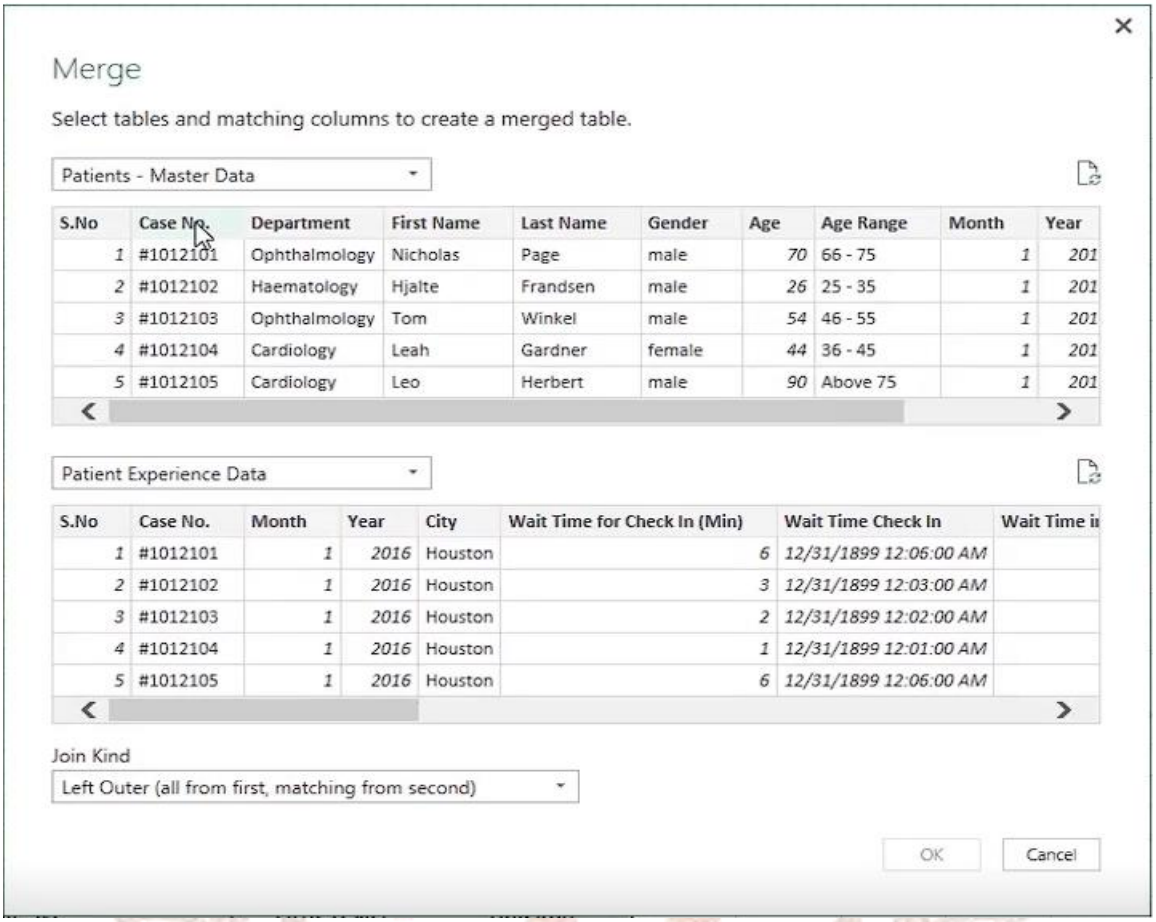


Fig 2: Step 2 Selecting Merge Data

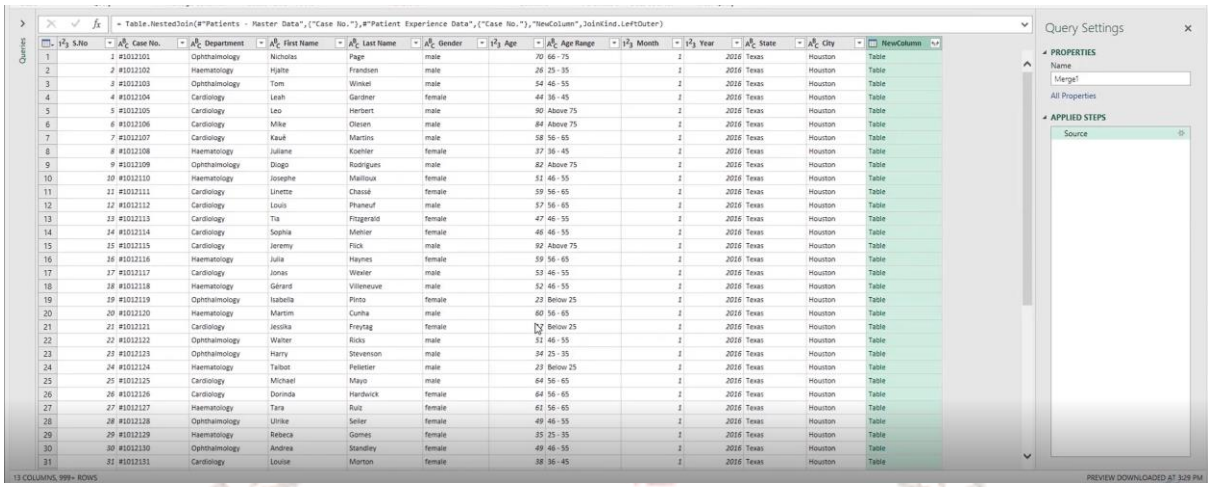


S.No	Case No.	Department	First Name	Last Name	Gender	Age	Age Range	Month	Year
1	#1012101	Ophthalmology	Nicholas	Page	male	70	66 - 75	1	201
2	#1012102	Haematology	Hjalte	Frandsen	male	26	25 - 35	1	201
3	#1012103	Ophthalmology	Tom	Winkel	male	54	46 - 55	1	201
4	#1012104	Cardiology	Leah	Gardner	female	44	36 - 45	1	201
5	#1012105	Cardiology	Leo	Herbert	male	90	Above 75	1	201

S.No	Case No.	Month	Year	City	Wait Time for Check In (Min)	Wait Time Check In	Wait Time i
1	#1012101	1	2016	Houston	6	12/31/1899 12:06:00 AM	
2	#1012102	1	2016	Houston	3	12/31/1899 12:03:00 AM	
3	#1012103	1	2016	Houston	2	12/31/1899 12:02:00 AM	
4	#1012104	1	2016	Houston	1	12/31/1899 12:01:00 AM	
5	#1012105	1	2016	Houston	6	12/31/1899 12:06:00 AM	

Fig 3: Step 3 Merging Data Join Element

We need to choose the left outer as a join kind and then have to select the common join element as shown in figure 3. In this case the common column or field that need to join on is the case number. After selecting the common join element, we need to click on 'ok'. The data editor window will open which shows the columns from both the data sheets appearing in the single window as shown in figure 4.



	S.No	Case No.	Department	First Name	Last Name	Gender	Age	Age Range	Month	Year	State	City	NewColumn
1	1	#1012101	Ophthalmology	Nicholas	Page	male	70	66 - 75	1	2016	Texas	Houston	Table
2	2	#1012102	Haematology	Hjalte	Frandsen	male	26	25 - 35	1	2016	Texas	Houston	Table
3	3	#1012103	Ophthalmology	Tom	Winkel	male	54	46 - 55	1	2016	Texas	Houston	Table
4	4	#1012104	Cardiology	Leah	Gardner	female	44	36 - 45	1	2016	Texas	Houston	Table
5	5	#1012105	Cardiology	Leo	Herbert	male	90	Above 75	1	2016	Texas	Houston	Table
6	6	#1012106	Cardiology	Mike	Olesen	male	84	Above 75	1	2016	Texas	Houston	Table
7	7	#1012107	Cardiology	Kauê	Martins	male	58	56 - 65	1	2016	Texas	Houston	Table
8	8	#1012108	Haematology	Juliane	Koehler	female	37	36 - 45	1	2016	Texas	Houston	Table
9	9	#1012109	Ophthalmology	Diogo	Rodrigues	male	82	Above 75	1	2016	Texas	Houston	Table
10	10	#1012110	Haematology	Josephe	Mailloux	female	51	46 - 55	1	2016	Texas	Houston	Table
11	11	#1012111	Cardiology	Linette	Chassé	female	59	56 - 65	1	2016	Texas	Houston	Table
12	12	#1012112	Cardiology	Louis	Phaneuf	male	57	56 - 65	1	2016	Texas	Houston	Table
13	13	#1012113	Cardiology	Tia	Fitzgerald	female	47	46 - 55	1	2016	Texas	Houston	Table
14	14	#1012114	Cardiology	Sophia	Mehler	female	46	46 - 55	1	2016	Texas	Houston	Table
15	15	#1012115	Cardiology	Jeremy	Flick	male	92	Above 75	1	2016	Texas	Houston	Table
16	16	#1012116	Haematology	Julia	Haynes	female	59	56 - 65	1	2016	Texas	Houston	Table
17	17	#1012117	Cardiology	Jonas	Wexler	male	53	46 - 55	1	2016	Texas	Houston	Table
18	18	#1012118	Haematology	Gérard	Villeneuve	male	52	46 - 55	1	2016	Texas	Houston	Table
19	19	#1012119	Ophthalmology	Isabella	Pinto	female	23	Below 25	1	2016	Texas	Houston	Table
20	20	#1012120	Haematology	Martim	Cunha	male	60	56 - 65	1	2016	Texas	Houston	Table
21	21	#1012121	Cardiology	Jesslka	Freytag	female	19	Below 25	1	2016	Texas	Houston	Table
22	22	#1012122	Ophthalmology	Walter	Ricks	male	51	46 - 55	1	2016	Texas	Houston	Table
23	23	#1012123	Ophthalmology	Harry	Stevenson	male	34	25 - 35	1	2016	Texas	Houston	Table
24	24	#1012124	Haematology	Talbot	Pelletier	male	23	Below 25	1	2016	Texas	Houston	Table
25	25	#1012125	Cardiology	Michael	Mayo	male	64	56 - 65	1	2016	Texas	Houston	Table
26	26	#1012126	Cardiology	Dorinda	Hardwick	female	64	56 - 65	1	2016	Texas	Houston	Table
27	27	#1012127	Haematology	Tara	Ruiz	female	61	56 - 65	1	2016	Texas	Houston	Table
28	28	#1012128	Ophthalmology	Ulrike	Seiler	female	49	46 - 55	1	2016	Texas	Houston	Table
29	29	#1012129	Haematology	Rebeca	Gomes	female	35	25 - 35	1	2016	Texas	Houston	Table
30	30	#1012130	Ophthalmology	Andrea	Standley	female	49	46 - 55	1	2016	Texas	Houston	Table
31	31	#1012131	Cardiology	Louise	Morton	female	38	36 - 45	1	2016	Texas	Houston	Table

Fig 4: Step 4 Merged Data Sheet

The columns from the second sheet are concatenated and are created as a single column called the new column. We need to click on the expand icon and then select the data fields that are required for the new merge data source. Since there will be repeating data columns we need to choose only the required data, in this case, our required data would be 'wait time for check in', 'wait time in waiting room', 'wait time for physician', 'wait time at check out', 'total wait time' and 'patient satisfaction rating' as shown in figure 5. After this we need to click ok to display to the selected data as shown in figure 6.

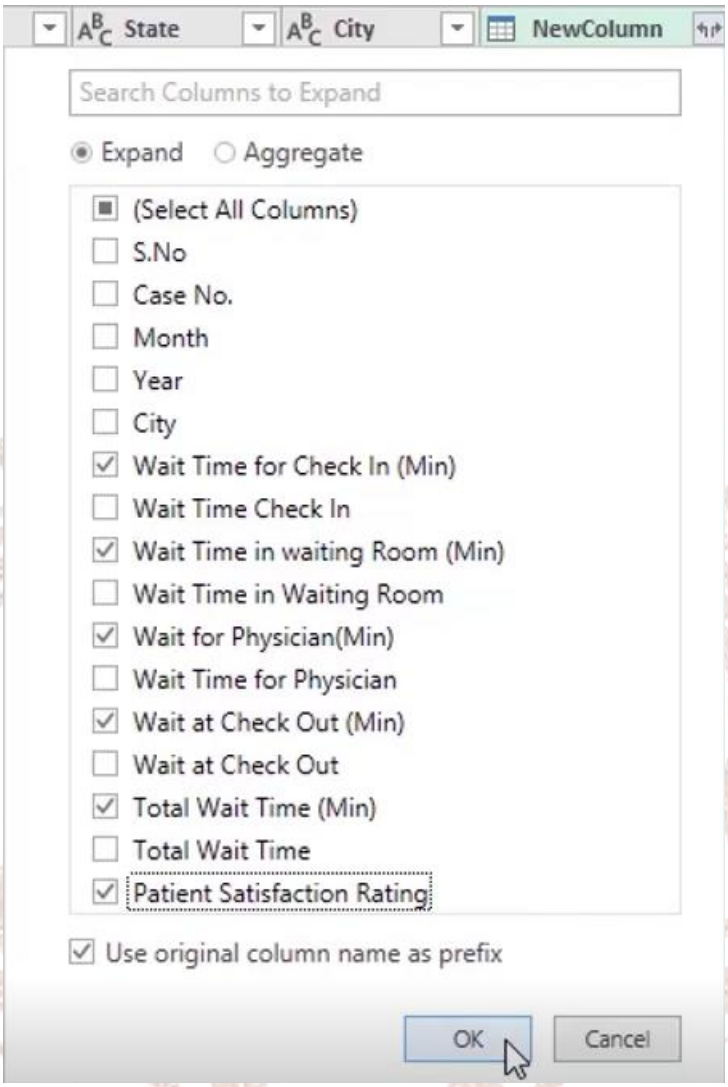


Fig 5: Data Fields to Display

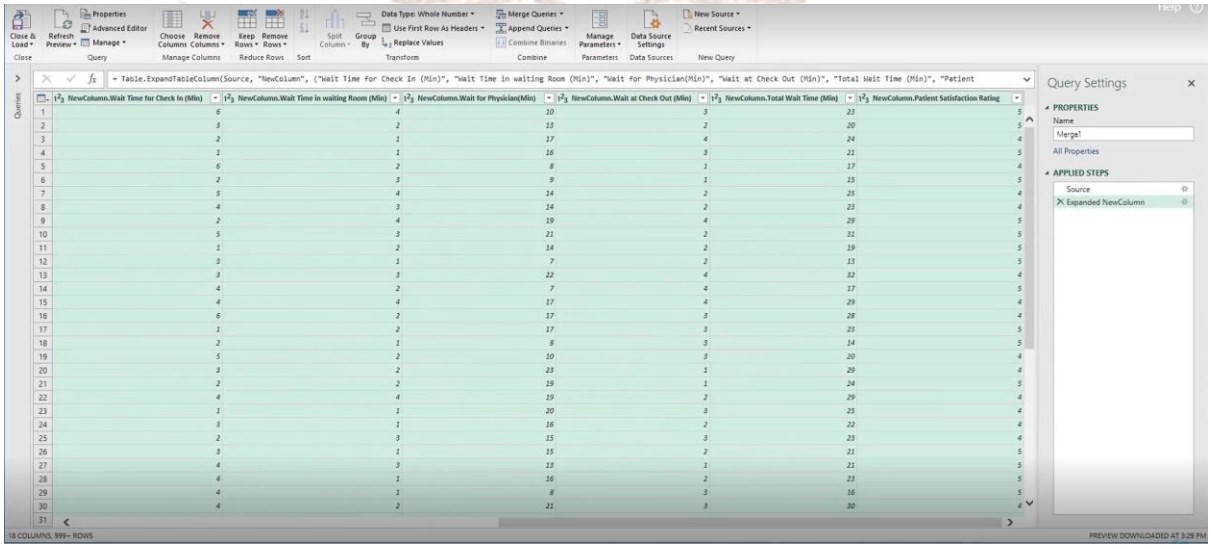


Fig 6: Required Data Fields

Now we have required field from the data sources which we wanted to merge. After this we need to select the close and load option on the top left of the window. The merge data source will be loaded into the workbook query window which contains the data from both the master data file and the experience data file, as shown in figure 7.

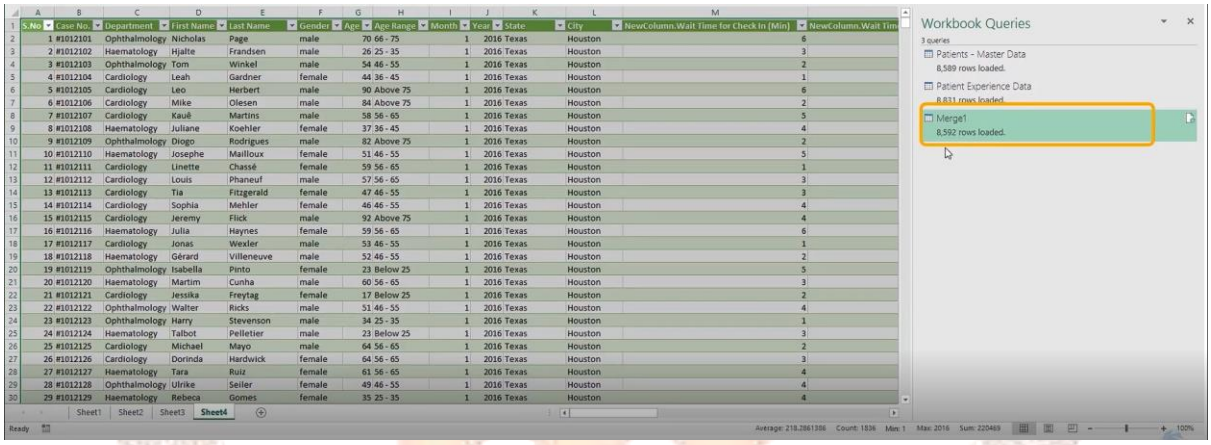


Fig 7: Loaded Merge Data Source

Once this is done, we have to right click on the merge file and rename it as needed. Here the merge data source named as hospital performance data which will be used to visualize the data as shown in figure 8.

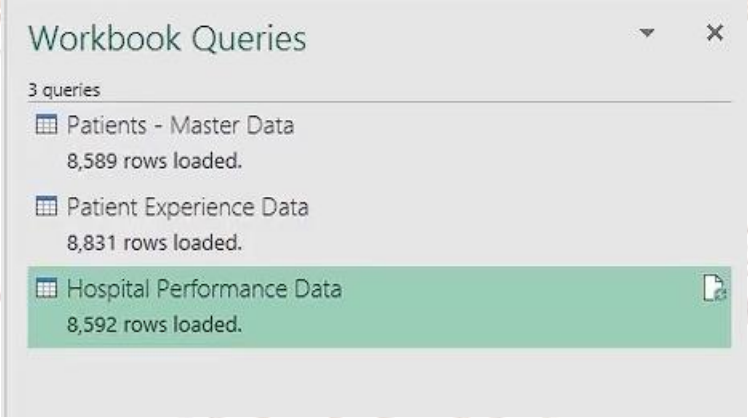


Fig 8: Renamed Merged Data Source

Editing and Transforming Data:

Now let's try to understand the event of modifying and transforming the data to match any required specifications.

First, we need to shape the dataset by deleting unnecessary columns, adding specific calculated columns, filtering data to our requirements. To do this, we need to right click on the merged data source which we renamed in figure 8 and click on edit. The query editor window will open where we can perform any of the required data modification or cleansing technique to prepare the data set for visualization.

Let's begin with renaming the existing columns, this can be done by either double clicking on the column name to change the name directly or right click on the column name and select the rename option to change the name of the column accordingly.

To delete or remove any existing columns, we need to right click on the column name and choose the option "remove" which will delete the column from the worksheet.

Note: the query editor keeps track of each transformation which has been performed under the applied steps window. All the transformations which have been applied makes up the new query. Note that none of these actions performed change the original source data, excel records each data that is performed and takes a snapshot of it and brings it back to the workbook.

To speed up the development of any complex data transformation processes we need to filter or remove data to reduce the sample dataset. We can use the 'Remove Rows' option in the top menu where we can do numerous operations like remove top rows, bottom rows, alternate rows, remove duplicates and errors which will help in removing certain records which will make the development of the data transformation process faster and easier.

Appending the Data:

In case if we have a requirement where the data is coming from two different data sources, but the data is similar, we can use the append option which will allow to combine rows from 2 or 3 different tables with similar data structures.

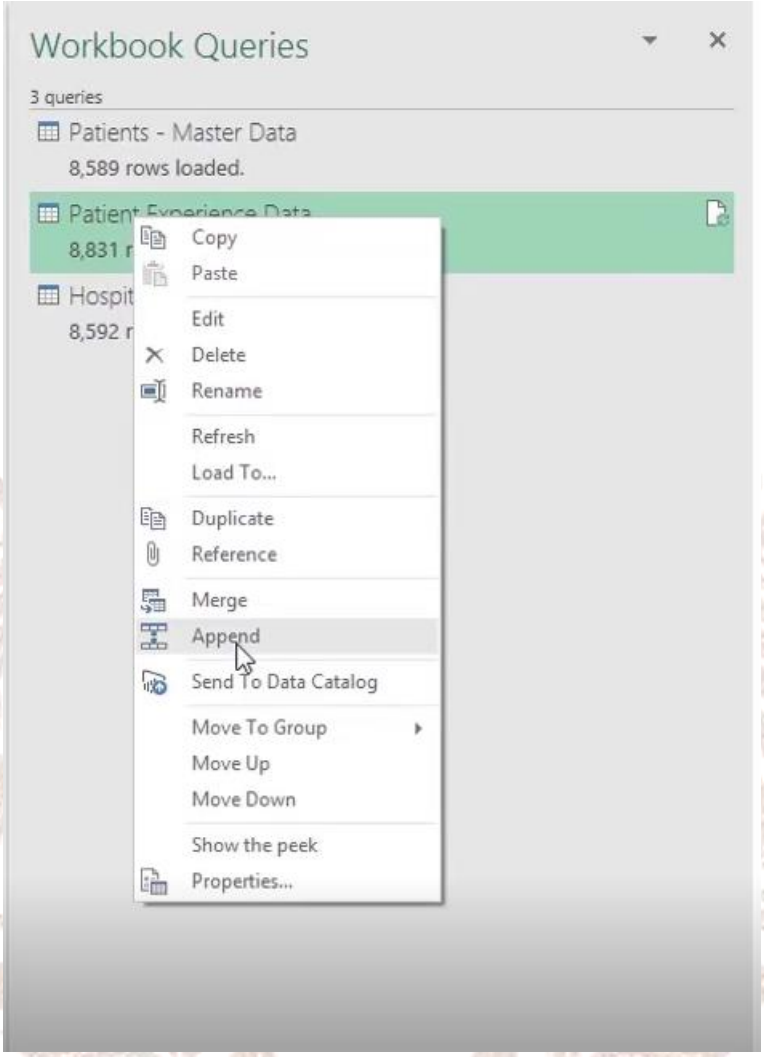


Fig 9: Append Option

We need to select the required data model, then right click to select the append option in the drop menu. This option helps in appending two or more different tables having same data structure. For example, if we have two or more rows appearing in one table and rest of rows with the same column structure appearing in another table. Now using append option, we can choose primary table which contain our first set of rows and second set of rows in second table, once they are chosen, we can press ok. This would append the rows from both the data sources.

Fig 10: Append Window

Data Manipulation with Power BI

Consider a scenario to sort patients by physician wait time. Here we considering the top 20 patients with the longest waiting time.

We need to sort the column in descending order to get the top 20 patients.

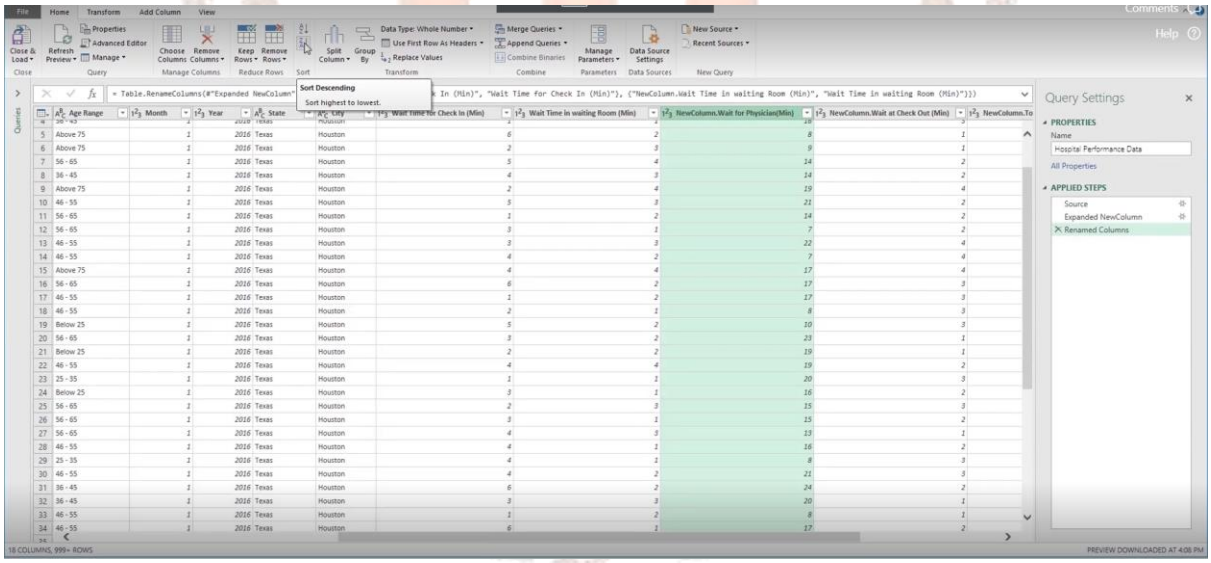


Fig 11: Sorting the Selected Column

The top 20 longest wait time patients are sorted and displayed as shown in figure 11. Once we have sorted data, we must select to the keep the top rows option and enter the number of rows required. For this above scenario we need 20 rows

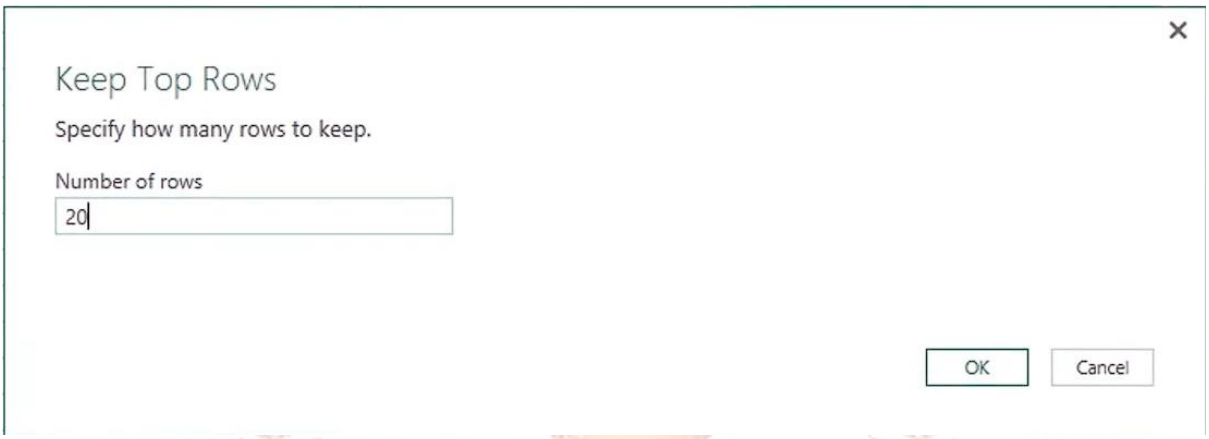


Fig 12: Keep Top Rows Window

The top 20 records for the highest physician wait time is displayed only, as shown in figure 13.

	NewColumn.Wait for Physician(Min)
1	55
4	55
3	55
1	55
3	55
2	53
2	53
4	53
1	52
4	52
2	52
3	51
1	50
3	50
4	48
2	48
4	48
4	48
4	48
2	48

Fig 13: 20 Sorted Rows

Note that all the steps performed will be added to the applied steps window in the query settings as shown in figure 14.

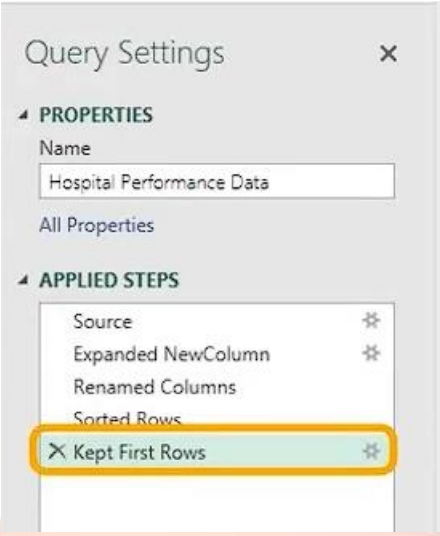


Fig 14: Applied Steps Window

Here we can revert to any previous version of the records by just deleting the step we need to undo. Here if we need to keep all the top end or bottom end records or even a specified range of records from the data sheet, we just need to delete the keep first rows step and all the data records will be displayed again.

Most datasets have duplicates resulting in poor data quality, so if we need to remove duplicate values from the records, we can remove the duplicates in the query editor

	S.No	Case No.
1	8803	#5212886
2	8813	#5212896
3	8808	#5212891
4	8797	#5212880
5	8818	#5212901
6	8802	#5212885
7	8812	#5212895
8	8819	#5212902
9	8815	#5212898
10	8807	#5212890
11	8823	#5212906
12	8825	#5212908
13	8805	#5212888
14	8809	#5212892
15	8796	#5212879
16	8820	#5212903
17	8828	#5212911
18	8804	#5212887
19	8828	#5212911
20	8800	#5212883
21	8795	#5212878
22	8821	#5212904
23	8806	#5212889
24	8826	#5212909
25	8826	#5212909
26	8798	#5212881
27	8827	#5212910
28	8827	#5212910
29	8799	#5212882
30	8816	#5212889

Fig 15: Data Records with Duplicates

Consider the case in figure 15 where there are duplicate records in the case number column. In the data records we have 37 rows and to remove the rows with duplicate records which in this case is 3 rows, we need to choose the case number column and then select the remove duplicate option in the remove rows feature in the top menu as displayed in figure 16.

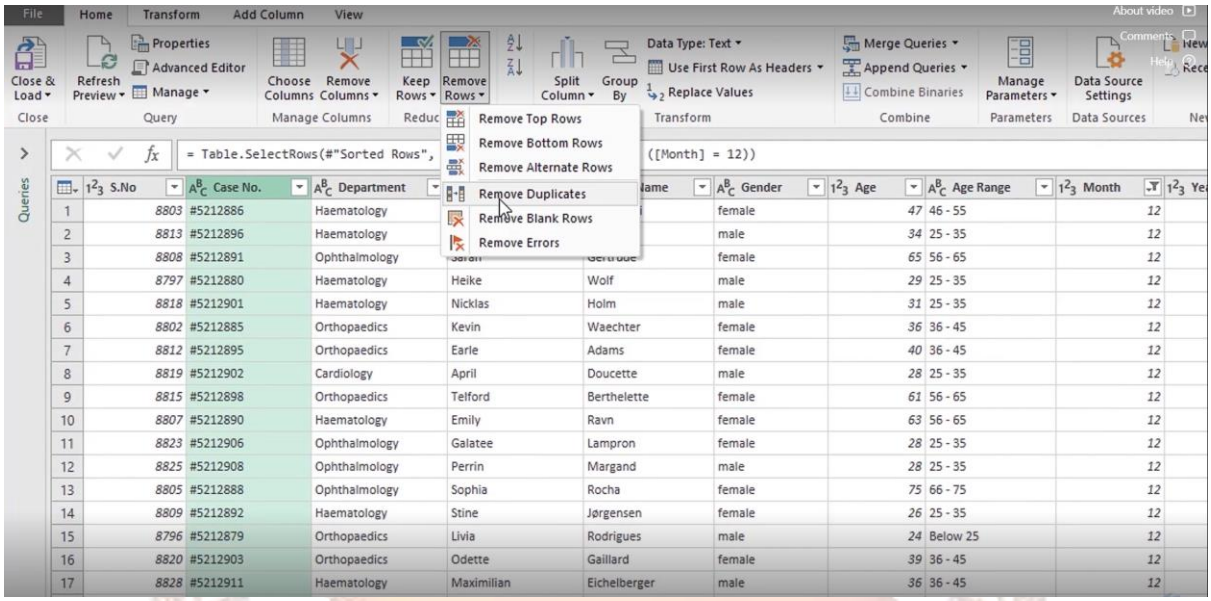


Fig 16: Delete Duplicate Option

Now we can see that the number of rows to be reduced to 34 and the 3 duplicate records are deleted. Similarly, if there are any column that consist of duplicate records which need to be deleted, select the column, and choose the remove duplicates option as shown in figure 16.

Filtering Data

Let's undo the changes made to the data records to get back the complete data record which is available. For this we have to delete the filtered rows option to obtain the original complete data.

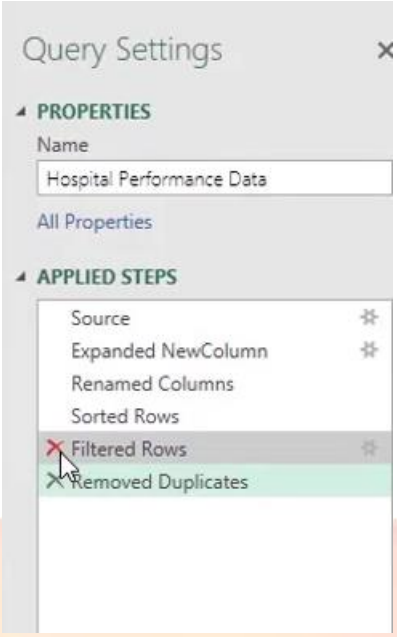


Fig 17: Obtaining the Complete Data

Once the complete data is obtained the process of limiting data by filtering can be done by choosing the filter option in the expand option adjacent to each column as shown in figure 18.

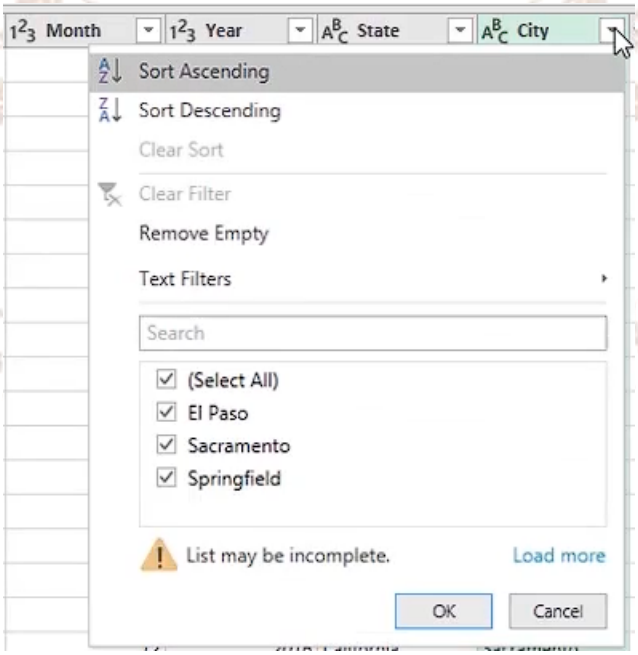


Fig 18: Filter Option

Here we can filter those records we do not need and display a limited set of data. For example, consider that the data from Boston and Chicago are not required, we can deselect the above-mentioned records and only display the required data.

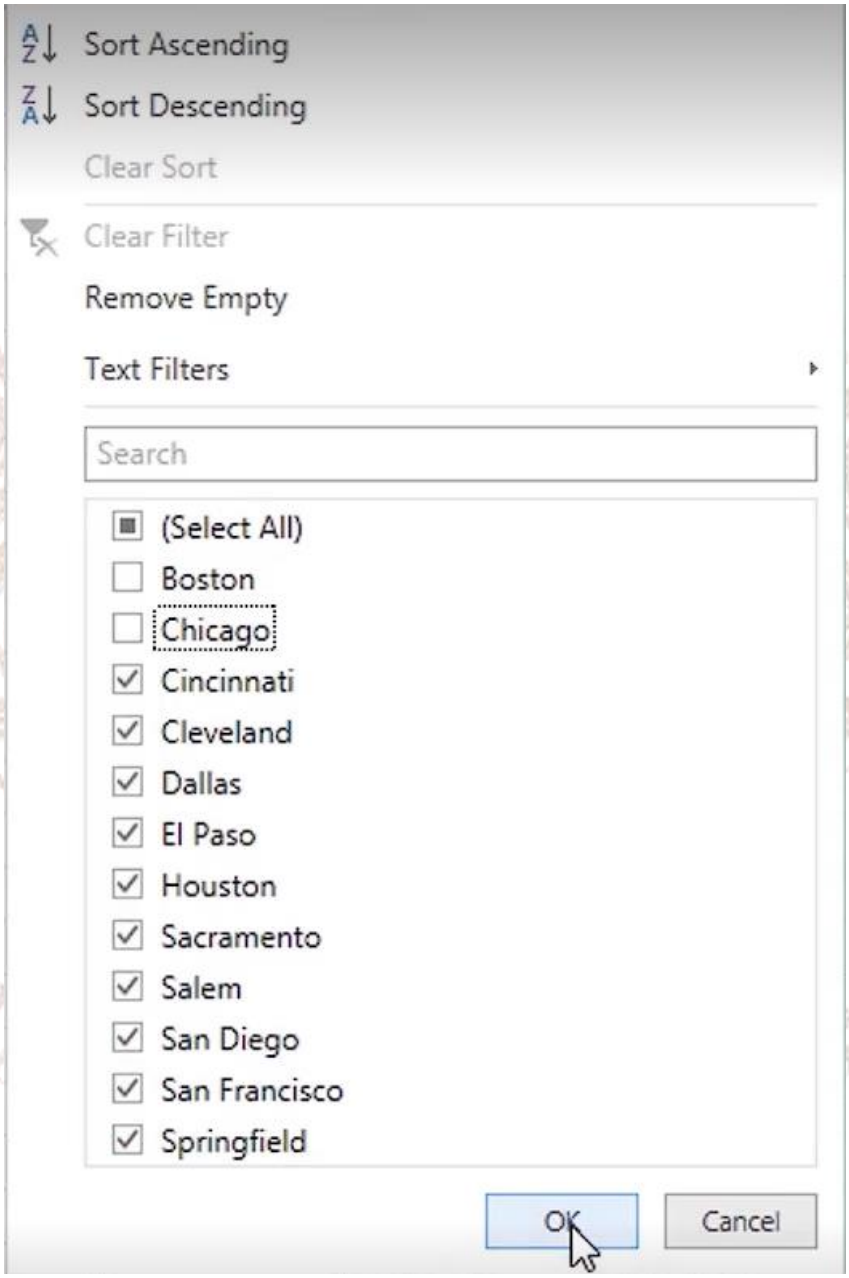


Fig 19: Filter Option

Once this option is applied the data will be filtered and the data from Boston and Chicago will not be displayed and only the remaining data will be obtained.

We can also perform this by using the text filter function.

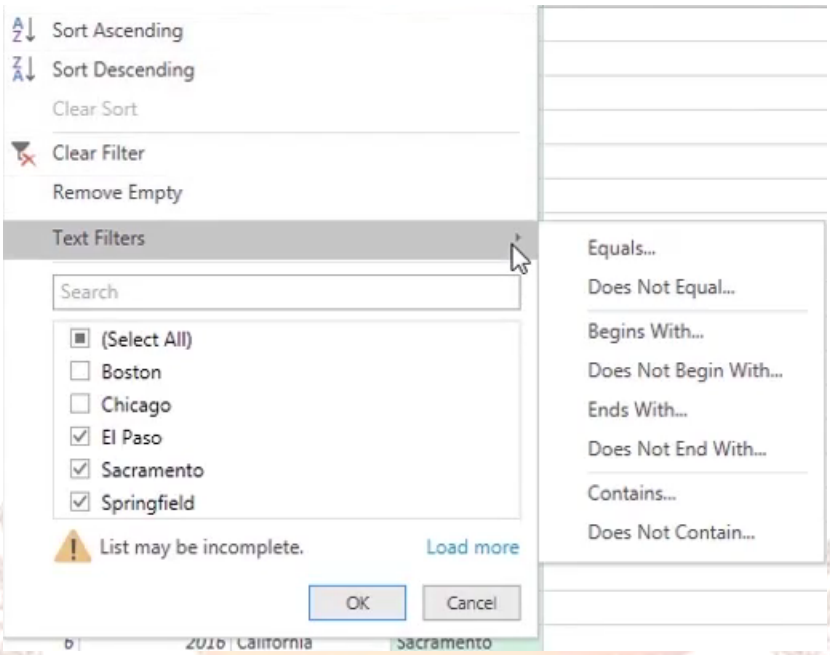


Fig 20: Text Filter Function

Figure 20 shows the text filter option where we can choose advance options to filter the data like equals, does not equal, begin with, does not begin with, end with etc.

When the data sheet consists of date records, we can filter the data using specific date option such as, next week, quarter, last year etc.

Inserting Custom Columns:

To insert custom column, say full name, in the dataset where we have the columns first name and last name. Here we need to concatenate the two columns to get the values for the full name column. To perform this operation, select the add custom column from the top menu which will open a add custom column window as shown in figure 21.

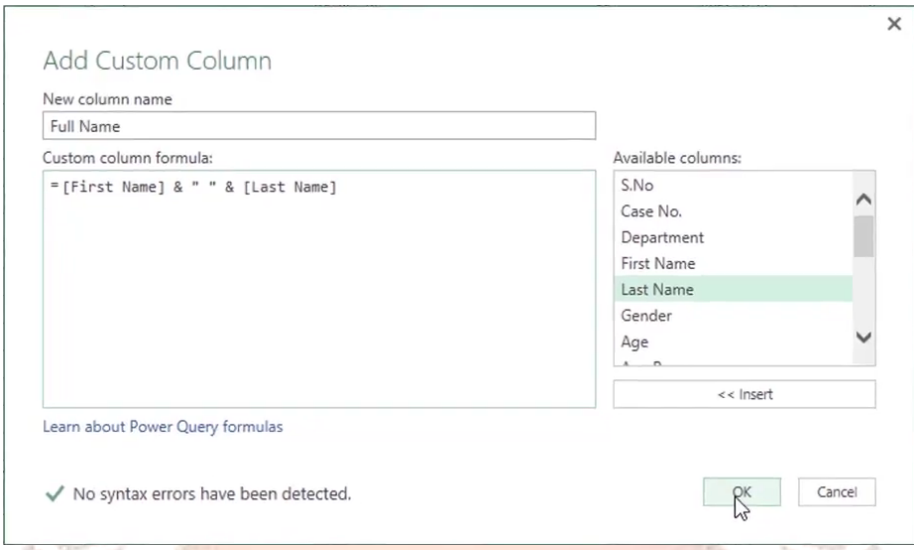


Fig 21: Adding Custom Column

Figure 21 shows the add custom column window, here we are adding the column full name, and, in the column, formula add the attributes first name and last name and concatenating the attributes with a space. After adding the attributes clicking ok will add the new custom column 'full name' to the work sheet as shown in figure 22.

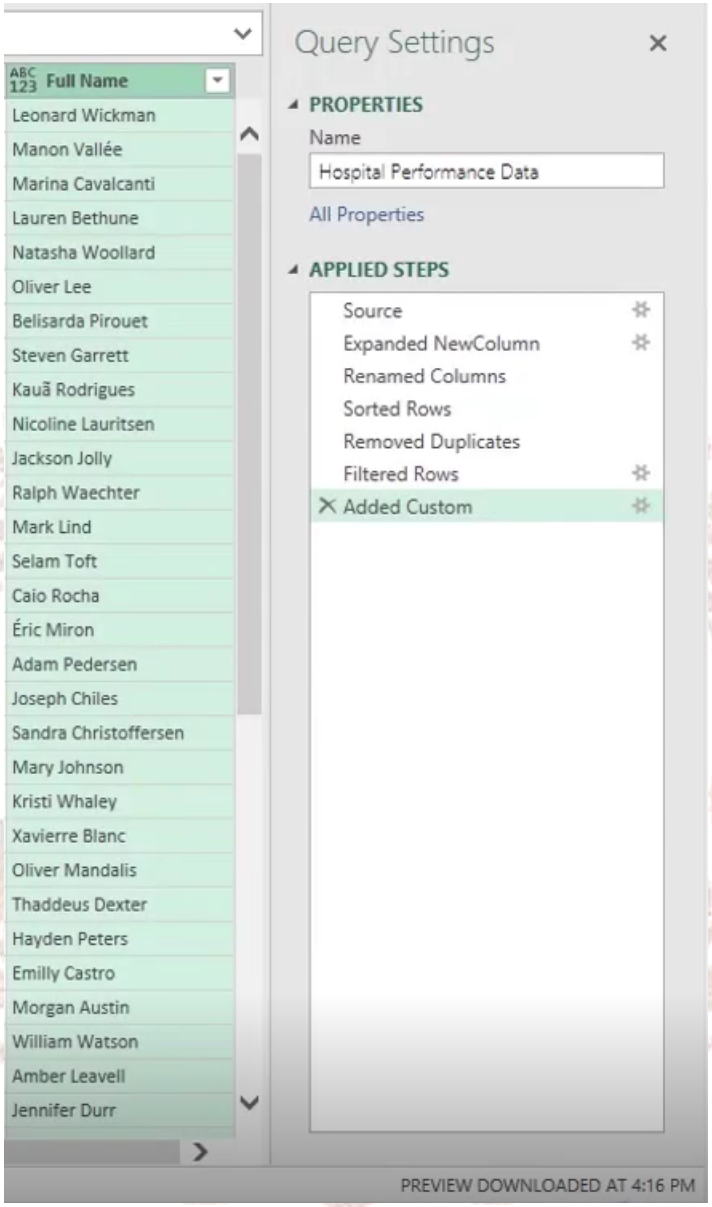


Fig 22: Custom Column

Cleanse and Modifying Data:

For the consumption of data, cleansing and modification of data is required to be done. The query editor automatically assigns the data type when the data is loaded from excel or other databases, so checking the data types is very important especially when we are looking for slicing and dicing of data. When the data is loaded from CSV or text files the query editor is unable to assign the data types, in this case the data type needs to be assigned manually. The query editor occasionally attempts to apply the data type on its own, this will add a new step in the navigator window known as 'changed type'.

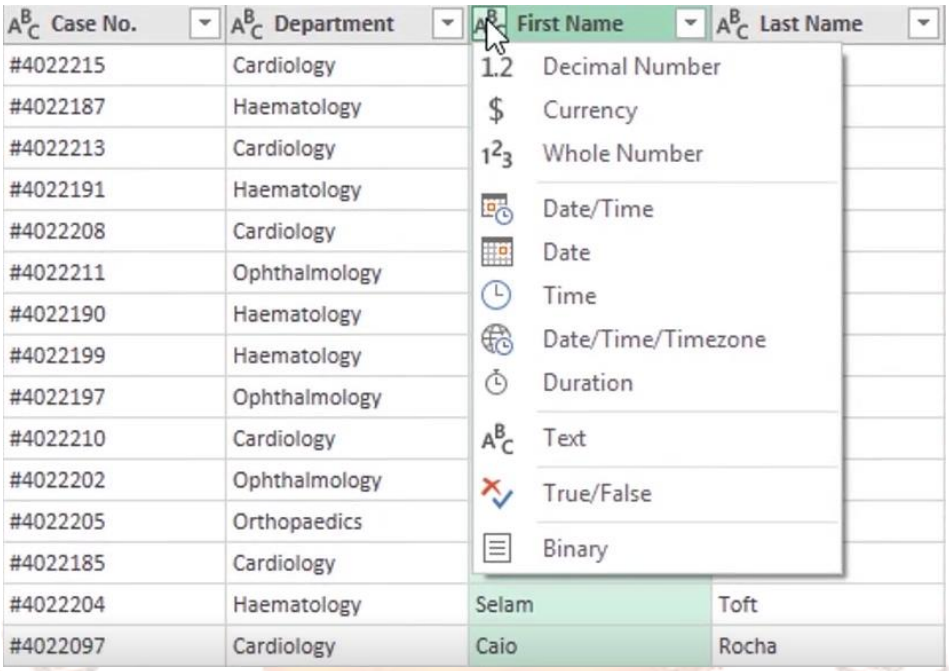


Fig 23: Changing Data Type Manually

We can change the data type manually as shown in the figure 23. Selecting the data and then doing right click on the data type will provide a dropdown menu from which we can choose the necessary data type. A pop up appears with all the data types options, be it date, binary, text, decimal, whole number etc.

Some data types cannot be applied to certain columns for example if we apply date data type to a text field then the entire column will show error data as displayed in figure 24.

Fig 24: Error Display

To restore the above error, we need to delete the change type step from the navigator window. This will restore the column to its original form.

Replacing The Column Values:

Replacing column values is useful when we are facing data quality issues. Consider data taken from two systems where, in system 1 we have female abbreviated as 'fe' and in system 2 female is stored as 'female' completely. To maintain data consistency, we need to store data of both the systems under the same name, to do this, we need to select the gender column and have to choose the "replace value" option in the top menu which will open the replace values window as shown in figure 25.

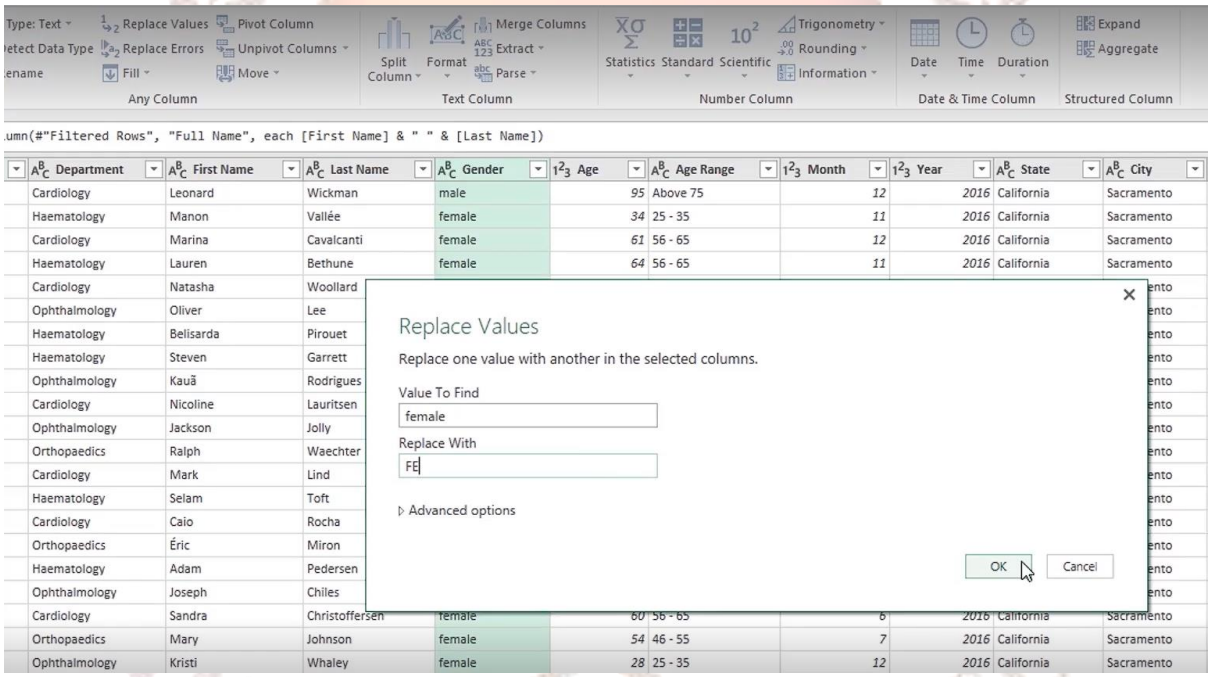


Fig 25: Replace Values Window

Here we need to enter the required value to be find and then have to enter the value we need to replace the searched values and then click ok. This will change all the female records in the column into 'FE' as displayed in figure 26.

Gender
male
FE
FE
FE
FE
male
FE
male
male
FE
male
male
male
FE
male
male
male
male
FE
FE
FE

Fig 26: Replaced Values for Gender Column

Changing data type and replacing values are the fundamental concepts required to manipulate and cleanse data in excel.

SELF-ASSESSMENT QUESTIONS – 1

1. Which one is not a function in MS Excel?
   a) SUM
   b) AVG
   c) MAX
   d) MIN
2. Concatenation of a text can be done using
   a) Apostrophe (‘)
   b) Exclamation (!)
   c) Hash (#)
   d) Ampersand (&)
3. If cells are entered the following values B1=72, B2=22, then in cell B3 , if the following formula is entered, Alt=. The result so called obtained is.
   a) 72
   b) 94
   c) 1584
   d) 3.2727
4. Remove Duplicates is the ribbon available in which tab?
   a) Home
   b) Insert
   c) Data
   d) Developer
5. Which of the following is not the component of Filtering data in excel?
   a) Number Filter
   b) Text Filter
   c) Date Filter
   d) Function Filter
6. Flash Fill can be used to separate first name and last name. Which one of the following is the wrong step to locate flash fill is _____.
   a) Home > Fill >Flash Fill
   b) Data>Flash Fill
   c) Using the short cut key Control + E
   d) View>Freeze Panes>Flash Fill

SELF-ASSESSMENT QUESTIONS – 1

7. Which one of the following is not the type of datatype?
   a) Binary
   b) Date
   c) Currency
   d) Integer
8. The _____automatically assigns the data type when the data is loaded from excel or other databases
9. When the data is loaded from _____the query editor is unable to assign the data types, in this case the data type needs to be assigned manually
10. Options like equals, does not equal, begin with, does not begin with, end with, contain, does not contain comes under which filter command
    a) Text Filter
    b) Number Filter
    c) Date Filter
    d) Function Filter

2. SUMMARY

In this topic we discussed,

- How to remove unnecessary columns and add specific calculated columns.
- To remove duplicates and infiltering the data
- To transform and cleanse data so to visualize it effectively.

3. TERMINAL QUESTIONS

Q1. Explain the need for appending the data?

Q2. Why deleting unnecessary columns, adding specific calculated columns, filtering data is important for editing and modifying data?

Q3. How data cleansing helps in data visualization?

Q4. Explain sorting and filter function in MS Excel?

Q5. What do you mean by data consistency? Explain this with an example

4. ANSWERS

Self-Assessment Questions

1. Option B, AVG is not a function instead average is the function
2. Option D, Ampersand (&)
3. Option B, 94
4. Option C, Data
5. Option D, Function Filter
6. Option D, View>Freeze Panes>Flash Fill
7. Option D, Integer
8. Query Editor
9. csv or text files
10. Option A, Text Filter

Answers-Terminal Questions

Ans1= A comprehensive data appending service usually includes a data normalization process, standardization of data, appending missing data, removing redundant data and setting up of data automation. Here is why your business should consider investing in data appending services.

- **Cleaner data**

Apart from verifying and completing your information, data appending can help you correct typos, update information (zip codes, place names or addresses) and check up on email/postal address errors. Data appending services can strengthen the validity of your mailing list.

- **Reduced time and effort**

Wrong phone numbers, bounced email addresses, returned mail can result in waste for your business, mainly in terms of time. In business, time is equivalent to money. Hence the more time you spend dealing with incorrect information, the lesser time you will have to focus on other activities of your business. With an updated mailing list that contains only the right information, your team of resourced can put their time to good use, without having to correct/update errors in the mailing list.

- **Access to more information**

The biggest benefit of data appending is access to information. With data appending services, your business can find information like gender, job roles, titles, birthdays, incomes and credit scores. Your business can find social media handles for Twitter, LinkedIn and Twitter. With access to more information, you will be able to send more effective marketing campaigns to your customers.

- **Better segmentation**

With access to more information, your business can customize its services through segmentation. Instead of only having access to name and age, data appending services can give you access to other critical information like income, which will help you in your marketing efforts. For instance, if you have a product for women in their mid-30s who make X amount of income per month, you will be able to find them through data appending.

- **Minimize cost**

Data appending services can help your business keep the cost down. With verified lists, you can save on the cost of research, recruiting staff, error correction and much more.

Ans2= First, we need to shape the dataset by deleting unnecessary columns, adding specific calculated columns, filtering data to our requirements. To do this, we need to right click on the merged data source which we renamed and click on edit. The query editor window will open where we can perform any of the required data modification or cleansing technique to prepare the data set for visualization.

Let's begin with renaming the existing columns, this can be done by either double clicking on the column name to change the name directly or right click on the column name and select the rename option to change the name of the column accordingly.

To delete or remove any existing columns, we need to right click on the column name and choose the option "remove" which will delete the column from the worksheet.

Note: the query editor keeps track of each transformation which has been performed under the applied steps window. All the transformations which have been applied makes up the new query. Note that none of these actions performed change the original source data, excel records each data that is performed and takes a snapshot of it and brings it back to the workbook.

To speed up the development of any complex data transformation processes we need to filter or remove data to reduce the sample dataset. We can use the 'Remove Rows' option in the top menu where we can do numerous operations like remove top rows, bottom rows, alternate rows, remove duplicates and errors which will help in removing certain records which will make the development of the data transformation process faster and easier.

Ans3= For the consumption of data, cleansing and modification of data is required to be done. The query editor automatically assigns the data type when the data is loaded from excel or other databases, so checking the data types is very important especially when we are looking for slicing and dicing of data. When the data is loaded from CSV or text files the query editor is unable to assign the data types, in this case the data type needs to be assigned manually. The query editor occasionally attempts to apply the data type on its own, this will add a new step in the navigator window known as 'changed type'.

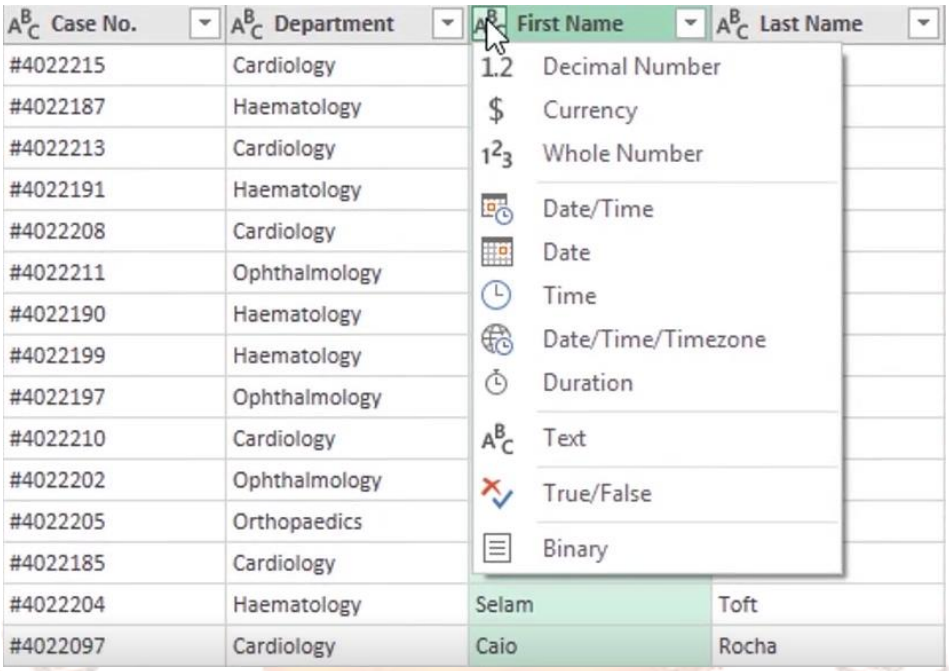


Fig 1: Changing Data Type Manually

We can change the data type manually as shown in the figure 1. Selecting the data and then doing right click on the data type will provide a dropdown menu from which we can choose the necessary data type. A pop up appears with all the data types options, be it date, binary, text, decimal, whole number etc.

Some data types cannot be applied to certain columns for example if we apply date data type to a text field then the entire column will show error data as displayed in figure 2

Fig 2: Error Display

To restore the above error, we need to delete the change type step from the navigator window. This will restore the column to its original form.

Ans4= There are many built-in Excel tools to help with data management and the sorting and filtering features are among the best. The filter tool gives you the ability to filter a column of data within a table to isolate the key components you need. The sorting tool allows you to sort by date, number, alphabetic order and more. In the following example, we will explore the usage of sorting and filtering and show some advanced sorting techniques.

Let's say you had the spreadsheet above and wanted to sort by price. This process is fairly simple. You can either highlight the whole column or even click on the first cell in the column to get started. Then you will:

- Right click to open the menu
- Go down to the Sort option – when hovering over Sort the sub-menu will appear
- Click on Largest to Smallest
- Select Expand the selection
- Click OK

The whole table has now adjusted for the sorted column. Note: when the data in one column is related to the data in the remaining columns of the table, you want to select Expand the selection. This will ensure the data in that row carries over with sorted column data.

In addition to sorting, you may find that adding a filter allows you to better analyse your data. When data is filtered, only rows that meet the filter criteria will display and other rows will be hidden. With filtered data, you can then copy, format, print, etc., your data, without having to sort or move it first. To use a filter,

- Go to the Home ribbon, click the arrow below the Sort & Filtering icon in the Editing group and choose Filter.

  OR

- Go to the Data ribbon, and then click Filter in the Sort & Filter group.

You will notice that all of your column headings now have an arrow next to the heading name. Click on the arrow next to the heading with which you want to filter, and you will see a list of all the unique values in that column. Check the box next to the criteria you wish to match and click OK. Click on the arrow next to another heading to further filter the data.

To clear the filter, choose one of these options:

- Click on the Filter icon next to the heading and choose Clear Filter from "Name of Heading".
- Go to the Data ribbon and click the Clear icon in the Sort & Filter group.
- Go to the Home ribbon, click the arrow below the Sort & Filter icon in the Editing group and choose Clear.

Ans5= Data that is consistent refers to data that is formatted in a consistent way. This is great for people working with data because it means all the data can be handled in the same way.

Additionally, data consistency can also refer to data that is constant over time or some other relationship. For example, if you have a weather dataset, it would be considered consistent if there were no missing days/hours (depending on your metric).

In summary, data consistency is typically people in data science try to follow and keep in mind because it ultimately makes the process of using it much easier.

There are many instances when data needs to be transferred or processed according to business use case. After processing, there might be possibility that the values in different table have different values but for the same record. This calls for redesign with improved data consistency.

For Example, table with 3 columns with employee name, employee number, phone number. Employee Name - XYZ, Employee Number - 123 and Phone Number - 0000.785.563. You want to use this data in another table or maybe this time your source of data is different. In new table if you have the same employee details, Employee Name - XYZ, Employee Number -123 and Phone Number - 0000.785.563, it is consistent data and if you get different phone for same employee, then it is inconsistent data.

Hence, data must be validated according to certain rules using constraints, triggers, transaction, etc to get improved data consistency.